

Global Superstore

Power BI Student Build Guide

Project: Power BI Analytics - Executive Sales and Profitability Dashboard

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Before You Start

What you'll build: a 4-page interactive Power BI report - Executive Overview, Product and Profitability, Customer and Market, and Shipping and Operations - backed by a dedicated Date table and a library of DAX measures.

What you need: Power BI Desktop, with the Global Superstore dataset already loaded as a single table. This guide assumes that table is named Orders; rename it in the Fields pane if yours imported under a different name.

How to use this guide: work through the sections in order. Each visual lists its type, the exact fields to use, and the settings to apply. Read the short notes alongside the steps - they explain what each step is actually for, not just how to click it.

1. Build the Date Table

Why: Time intelligence such as prior-year comparison and year-to-date analysis requires a contiguous date table.

Steps:

1. Select Home → New Table.
2. Enter the DAX below.
3. Press Enter and verify that the Date table contains one row per date.
4. Create a relationship from Date[Date] to Orders[Order Date].
5. Mark Date as the date table using Modeling → Mark as Date Table.
6. Select Month Name → Column tools → Sort by column → Month Number.

```
Date =  
ADDCOLUMNS (  
    CALENDAR ( MIN ( Orders[Order Date] ), MAX ( Orders[Order Date] ) ),  
    "Year", YEAR ( [Date] ),  
    "Month Number", MONTH ( [Date] ),  
    "Month Name", FORMAT ( [Date], "MMMM" ),  
    "Year-Month", FORMAT ( [Date], "MMM yyyy" ),  
    "Quarter", "Q" & ROUNDUP ( MONTH ( [Date] ) / 3, 0 )  
)
```

2. Create the Measures Table

Create a small dedicated table named `_Measures` to keep every measure you build organized in one place, separate from the data columns.

Steps:

1. Home → Enter Data.
2. Name the table `_Measures`. No columns are needed.
3. Select Load.
4. Select `_Measures` in the Fields pane before writing each measure in Section 3, so they don't scatter across the data table.

3. Build the Core DAX Measures

Why measures, not just columns: You don't create a measure because DAX makes it possible. You create one because a business question requires it. Every measure below exists to answer a specific question, listed alongside it.

3.1 Executive Measures

Measure (DAX)	Business purpose
Total Sales = SUM(Orders[Sales])	Revenue
Total Profit = SUM(Orders[Profit])	Profit
Total Quantity = SUM(Orders[Quantity])	Units
Total Orders = DISTINCTCOUNT(Orders[Order ID])	Orders
Total Customers = DISTINCTCOUNT(Orders[Customer ID])	Customers
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)	Profit efficiency
Avg Order Value = DIVIDE([Total Sales], [Total Orders], 0)	Average order value

3.2 Time Intelligence

Needs the Date table and relationship from Section 1.

Measure (DAX)	Business purpose
Sales PY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))	Prior-year sales
Sales YoY % = DIVIDE([Total Sales] - [Sales PY], [Sales PY], 0)	Growth
Sales YTD = TOTALYTD([Total Sales], 'Date'[Date])	Year-to-date performance

3.3 Discount & Shipping

Measure (DAX)	Business purpose
Avg Discount % = AVERAGE(Orders[Discount])	Average discount
Shipping Cost % of Sales = DIVIDE(SUM(Orders[Shipping Cost]), [Total Sales], 0)	Shipping efficiency
Avg Shipping Delay (Days) = AVERAGEX(Orders, DATEDIFF(Orders[Order Date], Orders[Ship Date], DAY))	Shipping delay

4. Page 1 — Executive Overview

Story: How is the business performing overall?

4.1 KPI Cards

Business question: What are the top-line numbers for the business at a glance?

Steps:

1. Create six Card visuals across the top of the page.
2. Assign one measure to each: Total Sales, Total Profit, Profit Margin %, Total Orders, Total Customers, Avg Order Value.
3. Align the cards evenly across the canvas.
4. Use consistent title and label formatting across all six.
5. Apply a restrained background color and a subtle shadow.
6. Format currency measures as currency and percentage measures as percentages.
7. Keep each KPI label short (one or two words).

Check yourself: Of the six cards, which one represents scale? Which represents profitability? Which represents customer or order value?

4.2 Sales & Profit Trend

Business question: How have sales and profit changed over time?

Steps:

1. Insert a Line Chart.
2. Add Date[Year-Month] to the X-axis.
3. Add Total Sales and Total Profit to the Y-axis.
4. Sort the axis chronologically.
5. Turn markers on if the chart stays readable.
6. Use distinct but consistent colors for Sales and Profit.
7. Remove unnecessary visual clutter (gridlines, borders).
8. Add a descriptive title.

Why this matters: this chart isn't just two lines - it lets you compare how revenue is moving against how profitability is moving, which don't always move together.

4.3 Sales by Market

Business question: Which markets generate the most sales?

Steps:

1. Insert a Clustered Bar Chart.
2. Add Market to the Y-axis.
3. Add Total Sales to the X-axis.
4. Sort by Total Sales, descending.
5. Turn on data labels.
6. Remove unnecessary gridlines.
7. Add the title "Sales by Market."
8. Test the visual using the Year slicer once it's built (4.5).

Think about it: does the highest-sales market also have the highest profit? Keep this question in mind - Page 2's Profit Margin heatmap will help answer it.

4.4 Sales by Category

Steps:

1. Insert a Donut Chart.
2. Add Category to Legend.
3. Add Total Sales to Values.
4. Turn on data labels showing value or percentage, whichever stays readable.
5. Use the same category colors consistently across every page that references Category.
6. Add the title "Sales Contribution by Category."

4.5 Executive Slicers

Steps:

1. Add a slicer using Date[Year].
2. Add a Market slicer.
3. Add a Segment slicer.
4. Add a Category slicer.
5. Format all four consistently.
6. Go to View → Sync Slicers, and sync all four slicers across all four report pages.

5. Page 2 — Product & Profitability

Story: Where are we making money, and where might profitability be under pressure?

5.1 Category / Sub-Category Performance

Steps:

1. Insert a Clustered Bar Chart.
2. Add Category and Sub-Category to the axis using a hierarchy.
3. Add Total Sales or Total Profit as the value, depending on which story you want the chart to tell.
4. Enable drill-down on the visual.
5. Sort by the metric you chose.

Why this matters: a hierarchy lets you move from a broad category down to the sub-category level on the same chart, instead of building a separate visual for each level.

5.2 Discount vs Profit

Business question: Does higher discounting appear to be associated with weaker profitability?

Steps:

1. Insert a Scatter Chart.
2. Add Avg Discount % to the X-axis.
3. Add Total Profit to the Y-axis.
4. Add Product Name to Details (or a coarser grouping if the chart gets too crowded).
5. Add Total Sales to Size, if the chart stays readable.
6. Open the Analytics pane and add a trend line.

Why this matters: a trend line shows association, not proof of cause and effect - be careful not to overstate what the chart proves.

5.3 Top 10 Products

Steps:

1. Insert a Clustered Bar Chart.
2. Add Product Name to the Y-axis.
3. Add Total Sales to the X-axis.
4. Open the Filters pane for this visual.
5. Set the Product Name filter type to Top N.
6. Enter 10.
7. Set Total Sales as the ranking measure, then apply the filter.

8. Sort descending.
9. Turn on data labels.

Note: this uses Power BI's built-in Top N filter rather than a ranking DAX measure - it's the simpler tool for the job, and worth mastering before you reach for more complex ranking formulas.

5.4 Profit Margin Heatmap

Steps:

1. Insert a Matrix.
2. Add Region to Rows.
3. Add Category to Columns.
4. Add Profit Margin % to Values.
5. Apply conditional formatting → Background color.
6. Use a restrained diverging color scale so weak combinations stand out clearly.

Why this matters: this matrix answers where profitability is weak, which is a different - and often more useful - question than where sales are high.

6. Page 3 — Customer & Market

Story: Who contributes to the business, and where are the valuable customers located?

6.1 Sales by Segment

Steps:

1. Insert a Donut Chart.
2. Add Segment to Legend.
3. Add Total Sales to Values.
4. Turn on readable data labels.
5. Keep segment colors consistent with the rest of the report.

6.2 Top 10 Customers

Steps:

1. Insert a Clustered Bar Chart.
2. Add Customer Name to the Y-axis.
3. Add Total Sales to the X-axis.
4. Apply a Top N filter of 10, ranked by Total Sales.
5. Sort descending.
6. Turn on data labels.

Think about it: use this visual to discuss customer concentration - how much of total sales comes from just these 10 customers?

6.3 Customer & Market Performance Table

Steps:

1. Insert a Table visual.
2. Add Market, Total Customers, Total Orders, Total Sales, Total Profit, Profit Margin %, and Avg Order Value, in that order.
3. Apply conditional formatting to Sales or Profit where it's useful.
4. Format currency, percentage, and decimal fields appropriately.

Check yourself: a market can have many customers but still produce low value. Which metric in this table helps you tell customer count apart from customer value?

6.4 Region × Category Sales Matrix

Steps:

1. Insert a Matrix.
2. Add Region to Rows.
3. Add Category to Columns. 4.
- Add Total Sales to Values. 5.
- Apply conditional formatting.

Note: use this matrix to spot concentration and gaps across geography and category together.

7. Page 4 — Shipping & Operations

Story: How efficiently are orders being shipped?

7.1 Average Shipping Delay by Ship Mode

Steps:

1. Insert a Clustered Column Chart.
2. Add Ship Mode to the X-axis.
3. Add Avg Shipping Delay (Days) to the Y-axis.
4. Turn on data labels.
5. Sort from highest to lowest if that reads better.
6. Title it "Average Shipping Delay by Ship Mode."

7.2 Orders by Priority

Steps:

1. Insert a Column Chart.
2. Add Order Priority to the X-axis.
3. Add Total Orders to the Y-axis.
4. Sort into a business-friendly priority order if the underlying values support it.

Don't: assign colors like red, orange, yellow, and green to priority levels unless the business has actually defined what those colors should mean. Arbitrary color semantics can mislead more than they help.

7.3 Shipping Cost vs Sales by Ship Mode

Steps:

1. Insert a Line and Clustered Column Chart.
2. Add Ship Mode to the X-axis.
3. Add Total Sales to the Column Y-axis.
4. Add Shipping Cost % of Sales to the Line Y-axis.
5. Format the two value axes so the units are easy to tell apart.

Why this matters: absolute shipping cost and shipping cost as a percentage of sales answer two different questions - a market can have high shipping cost but a healthy percentage, or the reverse.

7.4 Shipping Delay by Market

Steps:

1. Insert a Clustered Bar Chart.
2. Add Market to the Y-axis.
3. Add Avg Shipping Delay (Days) to the X-axis.
4. Sort descending.
5. Add data labels.

Note: use this visual to identify markets that need operational investigation.

8. Apply the Dashboard Design System

Use one consistent design system across all four pages.

Element	Standard
Canvas	16:9
Font	Segoe UI / Aptos-style clean sans-serif
Background	Very light neutral
Primary visual color	Deep blue / teal family
Positive	Green family
Negative	Red family
Neutral	Muted grey / amber where appropriate
Cards	Light background, subtle shadow, rounded corners
Spacing	Consistent margins and alignment
Gridlines	Off unless they materially improve readability
Titles	Short, business-question-oriented

9. Add Navigation

Steps:

1. Create a small navigation strip or set of buttons across the top of each page.
2. Create four buttons: Executive Overview, Product & Profitability, Customer & Market, Shipping & Operations.
3. Set each button's Action to Page navigation, pointing at the matching page.
4. Keep button styling consistent across all four.
5. Test navigation in Reading view.

10. Add Tooltips & Check Interactions

Steps:

1. Create a new page and set it up as a tooltip page (Format page → Page information → Tooltip).
2. Add a small set of supporting KPIs or a compact breakdown to it.
3. On each visual where a richer hover would help, set Format → Tooltips → Type = Report page, and select your tooltip page.
4. Test the hover behavior on each assigned visual.

5. Review the visual interactions (Format → Edit interactions) so slicers and charts cross-filter each other logically.

11. Reading Your Dashboard: The Four-Level Framework

A dashboard isn't finished when the last chart is built. It's finished when you can explain what each chart means and what decision it supports. Use these four levels every time you look at a visual - your own, or someone else's.

Level	Ask yourself
1. Observation	What does the visual show?
2. Comparison	What is higher or lower, or changing?
3. Insight	Why does this matter to the business?
4. Recommendation	What should management investigate or do next?

Exercise: Pick one visual from each of your four pages and write one sentence for each of the four levels above. If you can't get to a Recommendation, you probably stopped at Observation.